PREDICTING COMPANY BANKRUPTCY

GROUP 11

Company Bankruptcy?

Bankruptcy occurs when a company finds itself in excessive debt; that is, when the value of its property, its assets, no longer covers the total amount of its debts

65%

of small businesses

fail within the first 10 years, with cash flow problems being the primary cause of bankruptcy

U.S. Bureau of Labor Statistics

Employee Impact

Job losses, unpaid wages, lost benefits, and psychological stress affect thousands of workers and their families

American Bankruptcy Institute

Economic Ripple Effect

Suppliers lose revenue, creditors face losses, local economies decline, and investor confidence diminishes across markets

Gallup

Reasons for Bankruptcy based on studies across all Industries

Poor Liquidity or Cash-Flow Problems

01. **Profit ≠ Cash**

A company can show profits but still struggle if cash isn't coming in quickly enough.

02. **Poor Working Capital**

Poor working capital management (high inventory, slow receivables) reduces available cash to pay urgent bills.

03. **Cash-Flow Insolvency**

As cash dries up, operations stall and the firm becomes cash flow insolvent, risking bankruptcy.

Result cash-flow insolvency → bankruptcy

Reasons for Bankruptcy based on studies across all Industries

High Leverage : Too Much Debt

A company can use debt to grow faster but if it borrows too much, it becomes highly leveraged.

$

Too Much Debt

Payments are bigger than what company earns

What Happens Next

- Company runs out of cash
- Banks lose trust
- Borrowing gets more expensive

End Result

Can't pay back loans

Company may go bankrupt (owes more than it owns)

Reasons for Bankruptcy based on studies across all Industries

Declining Sales or Operational Inefficiency

01. **Falling Sales**

Revenue drops due to competition, market loss, or outdated products, reducing the cash coming in.

02. **High Fixed Costs**

Rent, salaries, and machinery costs stay the same even when sales fall, pushing profit margins down.

03. **Ongoing Losses**

Inefficiencies raise costs, and repeated losses slowly drain retained earnings and cash reserves.

Result financial distress → bankruptcy

OBJECTIVES

01.

Understand the data using Data Visualisations and Predictive Modeling to Gain Insights on Company Bankruptcy

02.

Use Insights to Make Strategic Decisions for Early Bankruptcy Detection and improve Financial Stability

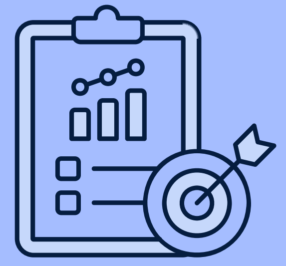
About the Dataset

95 Variables

6820 Observations

Liquidity & Leverage

- Quick Ratio
- Debt Ratio %
- Working Capital to Total Assets



Profitability & Performance Metrics

- **ROA(C) before interest and depreciation**
- **Operating Gross Margin**
- **Operating Profit Rate**

Growth Metrics

- **Realized Sales Gross Profit Growth Rate**
- **Operating Profit Growth Rate**
- **Total Asset Growth Rate**

Bankrupt Status

- **Bankrupt**

Data Cleaning & Feature Engineering

Standardize Variable Names
Standardized variable names and converted strings to lowercase for consistency

Correct Data Type
Assign correct data types for all variables to ensure data is consistent and ready for analysis

Duplicate Rows
Eliminated duplicate rows to remove redundant data

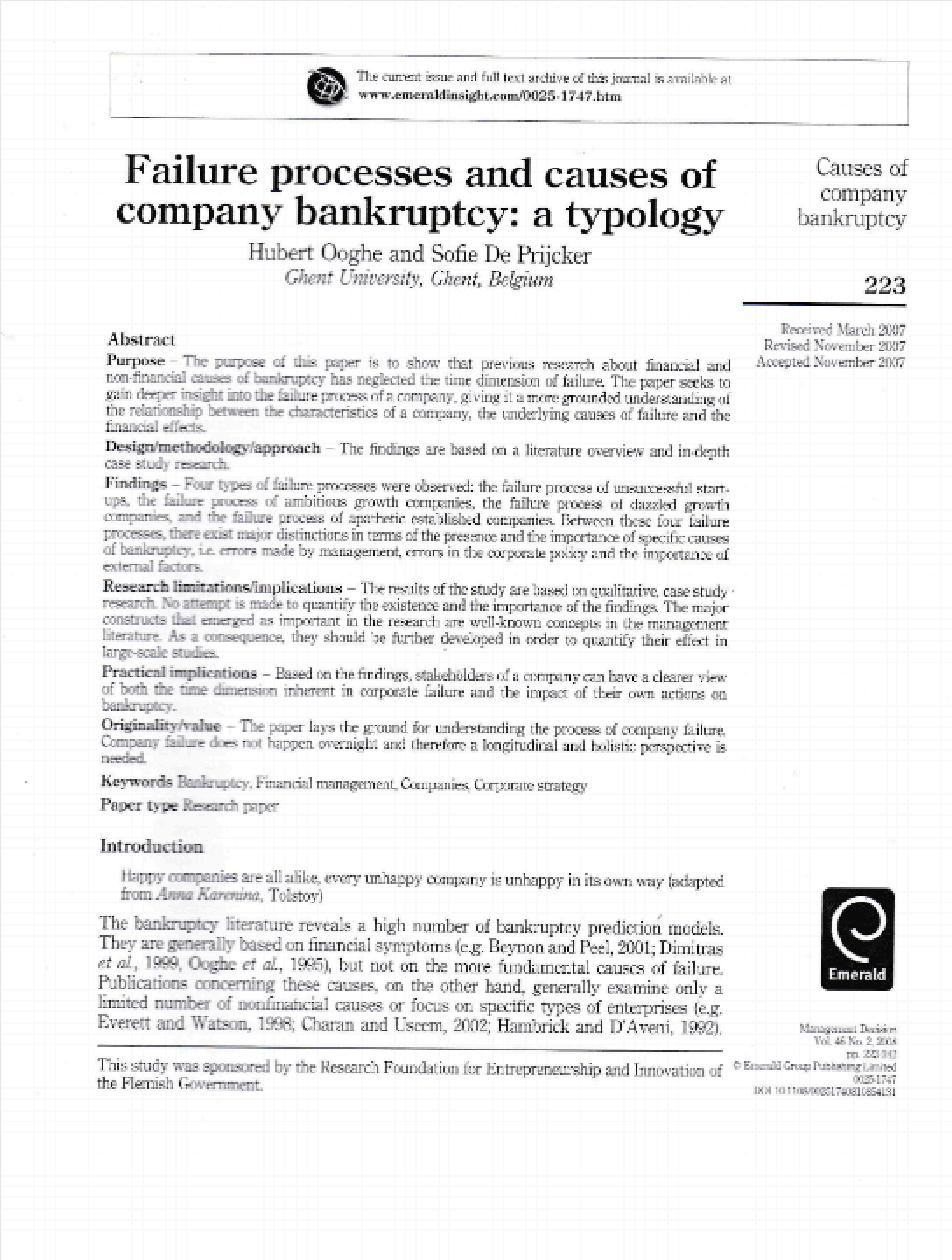
Outliers
Outliers existed but they were retained as they may be important in our analysis

No Missing Values
No Missing Values existed in any of the columns

Feature Reduction
Reduced features by removing highly correlated variables (correlation > 0.8) to eliminate redundancy and improve interpretability

Even after removing correlated variables, 70 features remained, making the analysis challenging

Feature Engineering - Using Domain Knowledge



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Causes of company bankruptcy

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Failure processes and causes of company bankruptcy: a typology

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Abstract

Purpose – The purpose of this paper is to show that previous research about financial and non-financial causes of bankruptcy has neglected the time dimension of failure. The paper seeks to gain deeper insight into the failure process of a company, giving it a more grounded understanding of the relationship between the characteristics of a company, the underlying causes of failure and the financial effects.

Design/methodology/approach – The findings are based on a literature overview and in-depth case study research.

Findings – Four types of failure processes were observed: the failure process of unsuccessful start-ups, the failure process of ambitious growth companies, the failure process of dazzled growth companies, and the failure process of apathetic established companies. Between these four failure processes, there exist major distinctions in terms of the presence and the importance of specific causes of bankruptcy, i.e. errors made by management, errors in the corporate policy and the importance of external factors.

Research limitations/implications – The results of the study are based on qualitative, case study research. No attempt is made to quantify the existence and the importance of the findings. The major constructs that emerged as important in the research are well-known concepts in the management literature. As a consequence, they should be further developed in order to quantify their effect in large-scale studies.

Practical implications – Based on the findings, stakeholders of a company can have a clearer view of both the time dimension inherent in corporate failure and the impact of their own actions on bankruptcy.

Originality/value – The paper lays the ground for understanding the process of company failure. Company failure does not happen overnight and therefore a longitudinal and holistic perspective is needed.

Keywords Bankruptcy, Financial management, Companies, Corporate strategy

Paper type Research paper

Introduction

Happy companies are all alike, every unhappy company is unhappy in its own way (adapted from *Anna Karenina*, Tolstoy)

The bankruptcy literature reveals a high number of bankruptcy prediction models. They are generally based on financial symptoms (e.g. Beynon and Peel, 2001; Dimitras *et al.*, 1999, Ooghe *et al.*, 1995), but not on the more fundamental causes of failure. Publications concerning these causes, on the other hand, generally examine only a limited number of nonfinancial causes or focus on specific types of enterprises (e.g. Everett and Watson, 1998; Charan and Useem, 2002; Hambrick and D'Aveni, 1992).

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Profit Margin (Net Income / Revenue)
Low or declining margin signals weak profitability, increasing bankruptcy risk.

Asset Turnover (Revenue / Total Assets)
Low turnover indicates inefficient asset use, leading to liquidity pressure

Return on Assets (Operating Income / Total Assets)
Consistently low ROA shows assets aren't generating enough income, signaling potential insolvency

Retained Earnings to Assets (Retained Earnings / Total Assets)
Low retained earnings imply high debt reliance, increasing vulnerability to financial distress

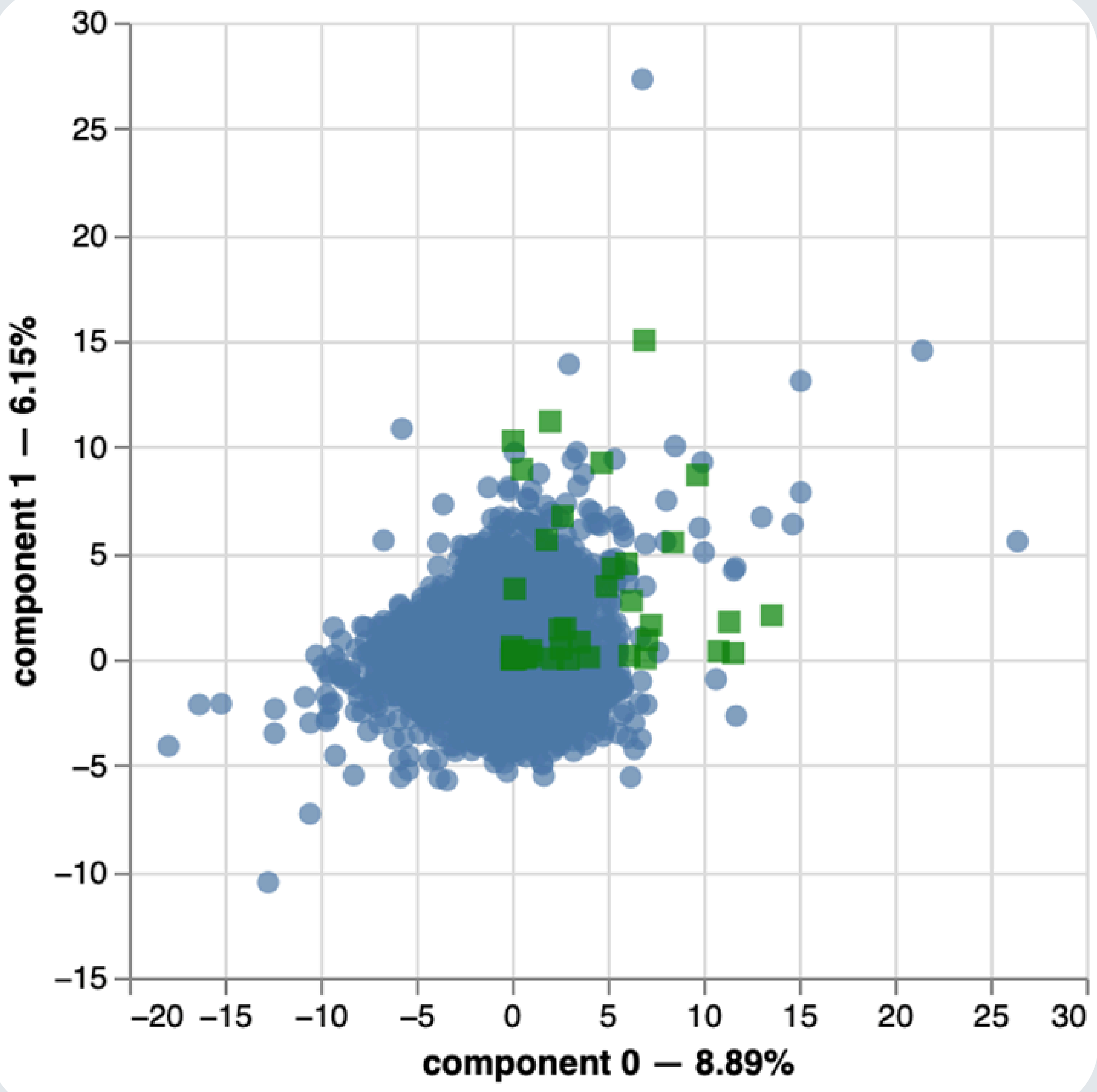
Working Capital to Assets (Current Assets – Current Liabilities) / Total Assets
Negative or low working capital reflects liquidity issues and difficulty meeting short-term obligations

P < 0.05
Mann Whitney U Test

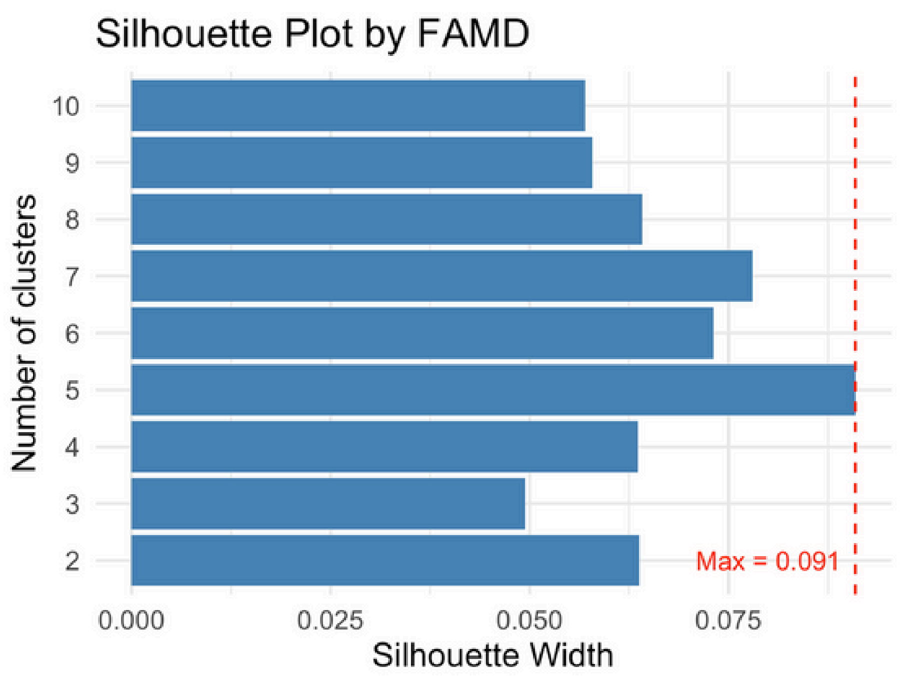
Significant Association between variables created and the bankruptcy status

BEFORE PROCEDDING TOWARDS MODEL FITTING

clusters are visible by the first component but, explains less that 15% of the variation



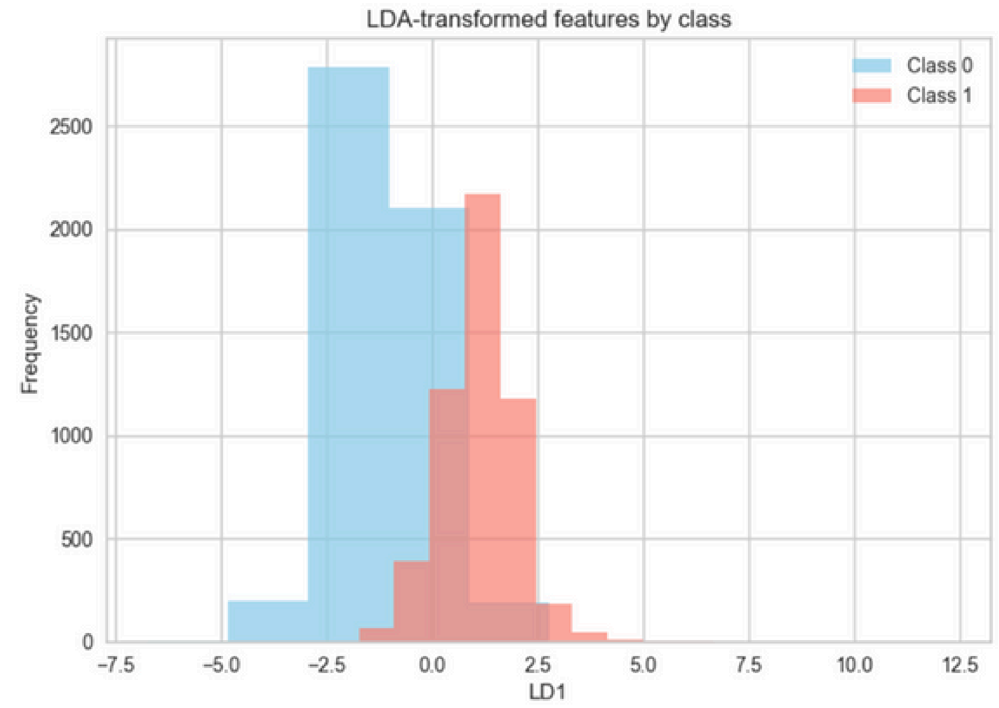
Get components that explain 70% of the variation



All the provided silhouette scores are very close to 0

K means Clustering → **No Significant Clusters**

Checking for linear seperability of classes using LDA



A simple logistic model would struggle here.

stacking non-linear tree models with a logistic layer on top

Liceria & Co.

MODEL BUILDING

Enables Data Driven Decision Making

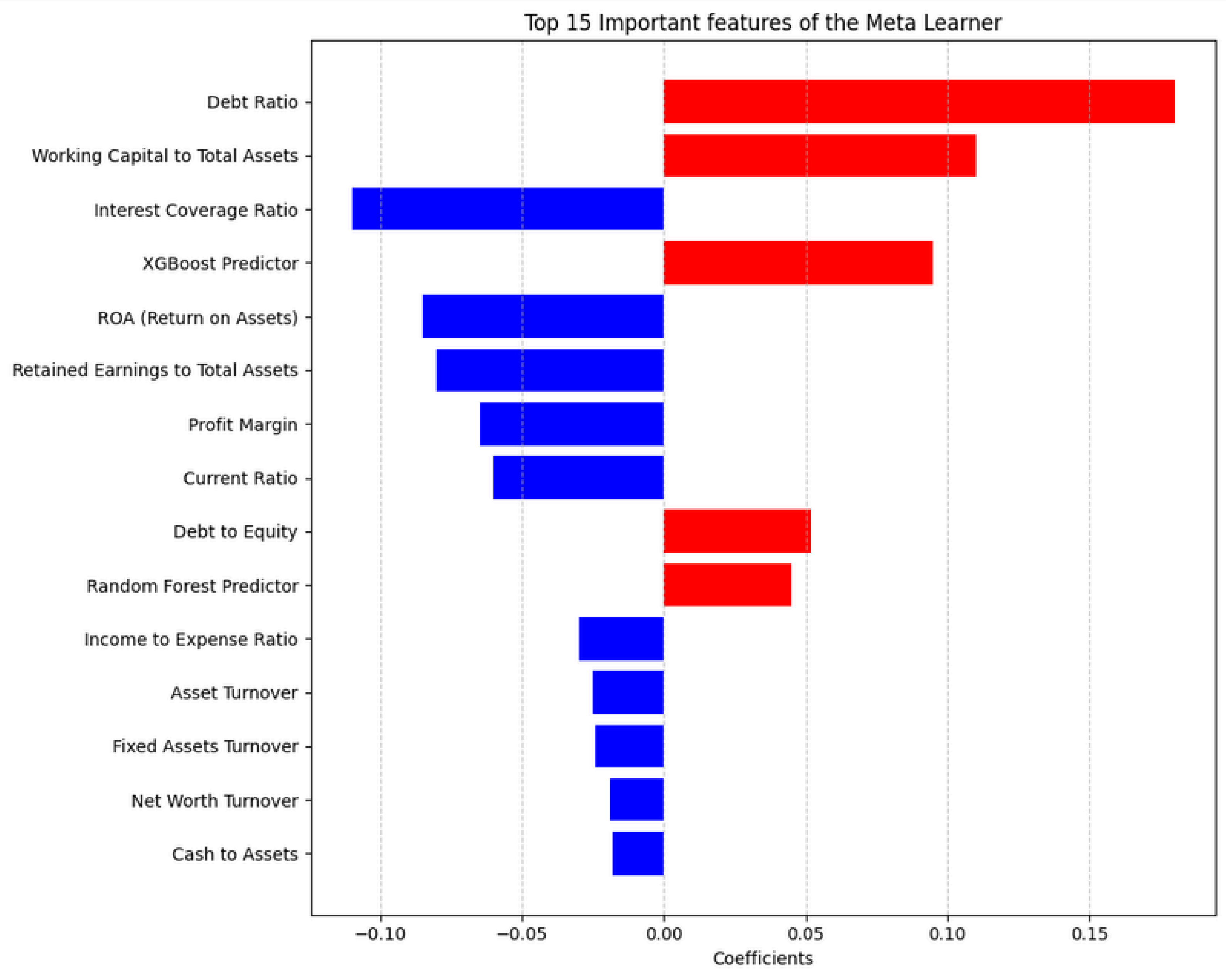
MODELS FOR PREDICTING COMPANY BANKRUPTCY

MEASURE	XGBOOST	EXTRA TREES CLASSIFIER	SVM	RANDOM FOREST	LOGISTIC REGRESSION	ADABOOST CLASSIFIER
TRAIN ACCURACY	0.97	0.94	0.80	0.90	0.67	0.85
TEST ACCURACY	0.93	0.89	0.71	0.86	0.68	0.79
PRECISION (0/1)	0.81 /0.69	0.87/0.84	0.52 / 0.46	0.86 / 0.86	0.60 / 0.54	0.62 /0.26
RECALL (0/1)	0.74 / 0.69	0.62 / 0.59	0.55 / 0.40	0.60 / 0.58	0.55 /0.45	0.77 / 0.59
F1 SCORE (0/1)	0.69 / 0.67	0.69 / 0.66	0.52 / 0.45	0.66 / 0.63	0.47 / 0.44	0.66 / 0.56

OVERFITTING ?

STACKED ENSEMBLE MODEL

01. Feature Importance

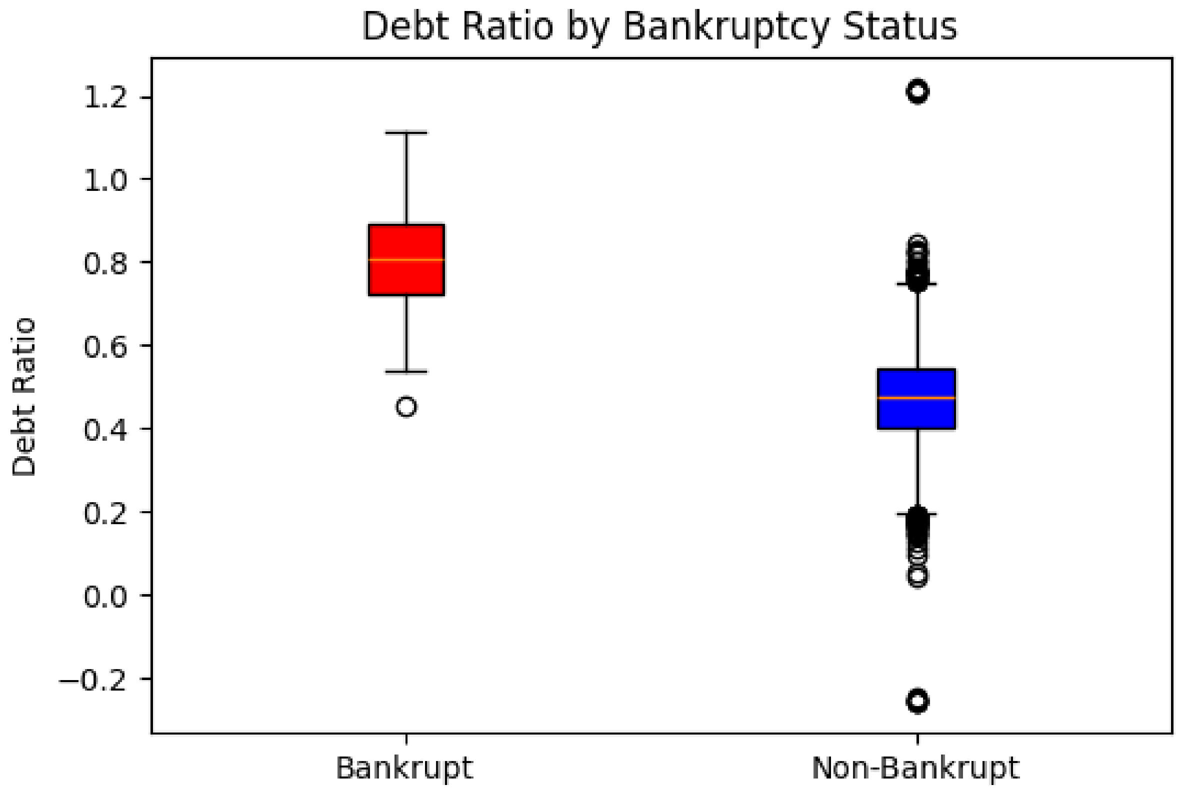


Metric	Value
Accuracy Train	0.921
Accuracy Test	0.913
precision (0/1)	0.81/0.75
Recall (0/1)	0.74/0.69
F1 score (0/1)	0.67/0.65

Most important Features

Debt Ratio
Working Capital to Total Assets
Interest Coverage Ratio
Retained Earnings to Total Assets
ROA (Return on Assets)
Profit Margin

Non-bankrupt companies take debt for growth
Bankrupt companies take debt for survival



$P < 0.05$

Mann Whitney U Test

Debt Ratio of Bankrupt Companies 2x more higher than non bankrupt companies

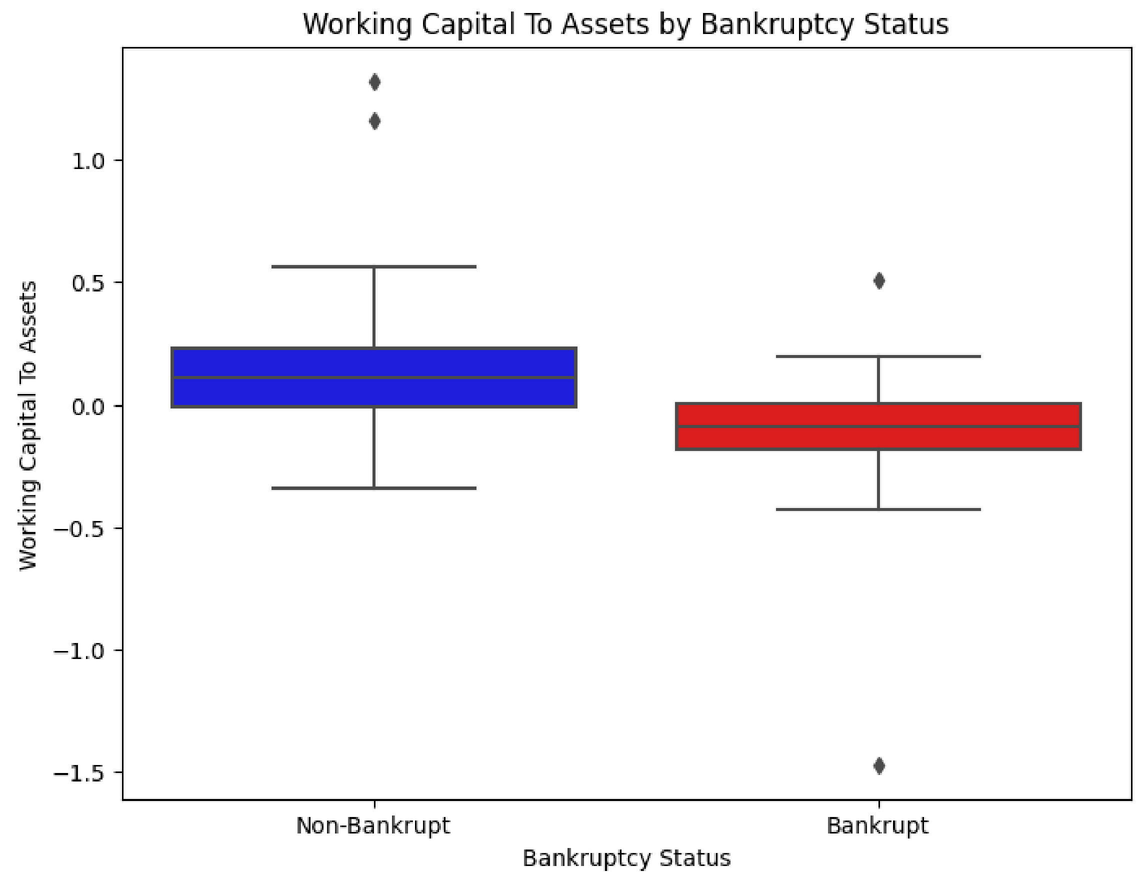
01. **Why Companies Borrow Debt**

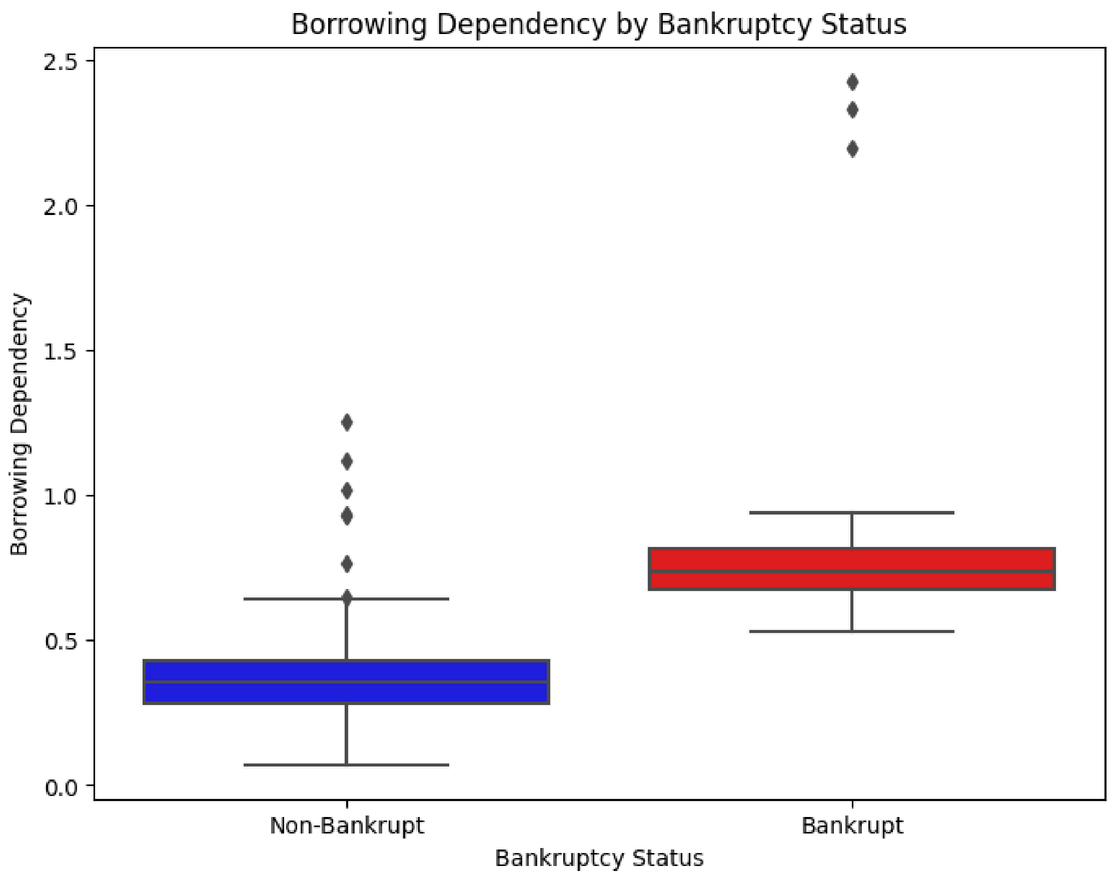
- To finance growth
- To manage cash flow
- To leverage opportunities

02. **The Issue with Debt of Bankrupt Companies**

- Their profits are too low, so they borrow to survive.
- They cannot generate enough internal funds
- They keep borrowing to cover losses, pushing debt ratios extremely high.

What are the early signs of bankruptcy?





working capital of bankrupt companies are significantly lower

Borrowing Dependency of Bankrupt Companies are 2x more higher

What are the early signs of bankruptcy?

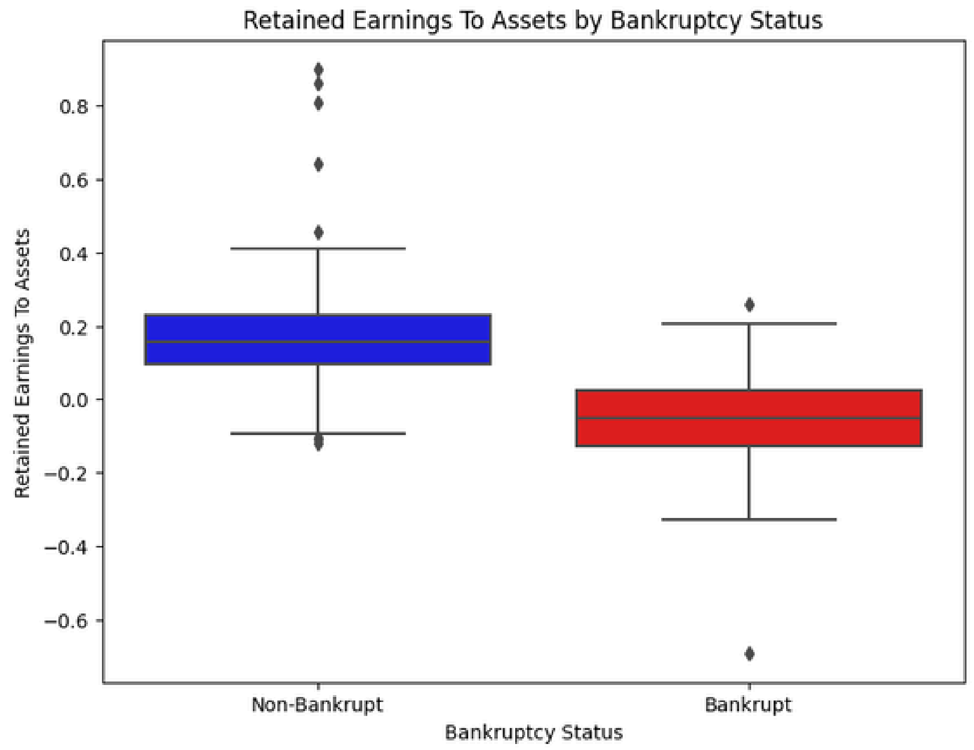
01. High debt drains cash through constant interest and repayment pressure

02. Liquidity shrinks as cash outflows increase — working capital and current ratio fall.

03. Low liquidity forces more borrowing, creating a cycle of debt → low cash → more debt

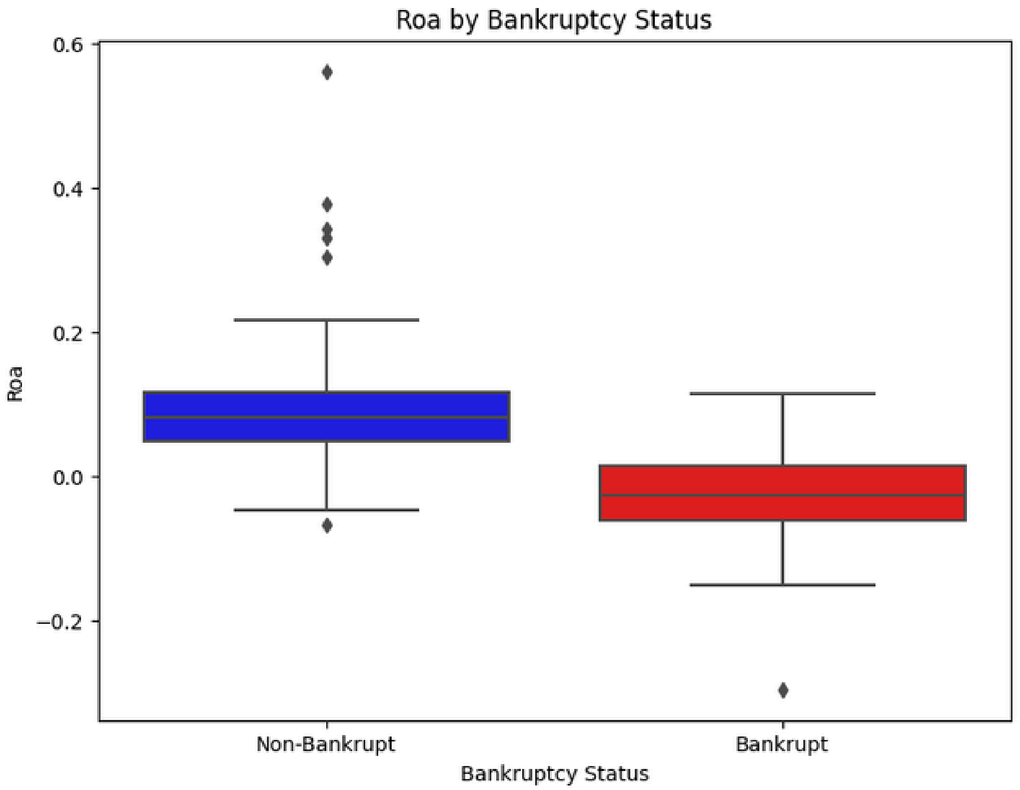
04. This self-feeding loop pushes the company toward financial distress

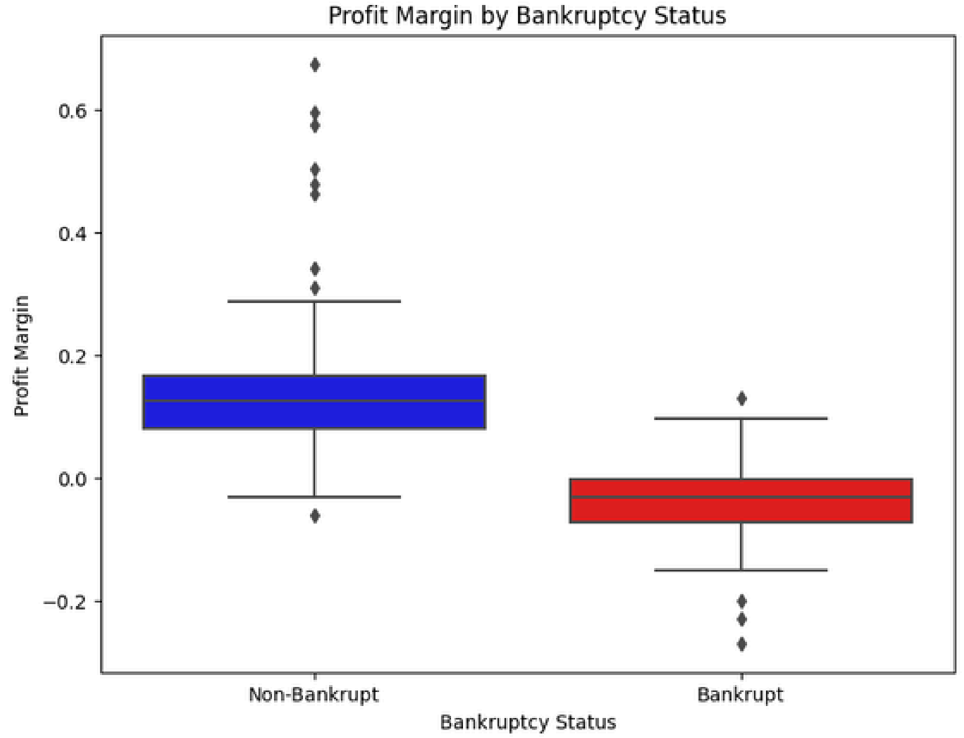
When liquidity tightens, the company loses internal financial strength



Out of everything the company owns (total assets), how much was built using its own accumulated profit?

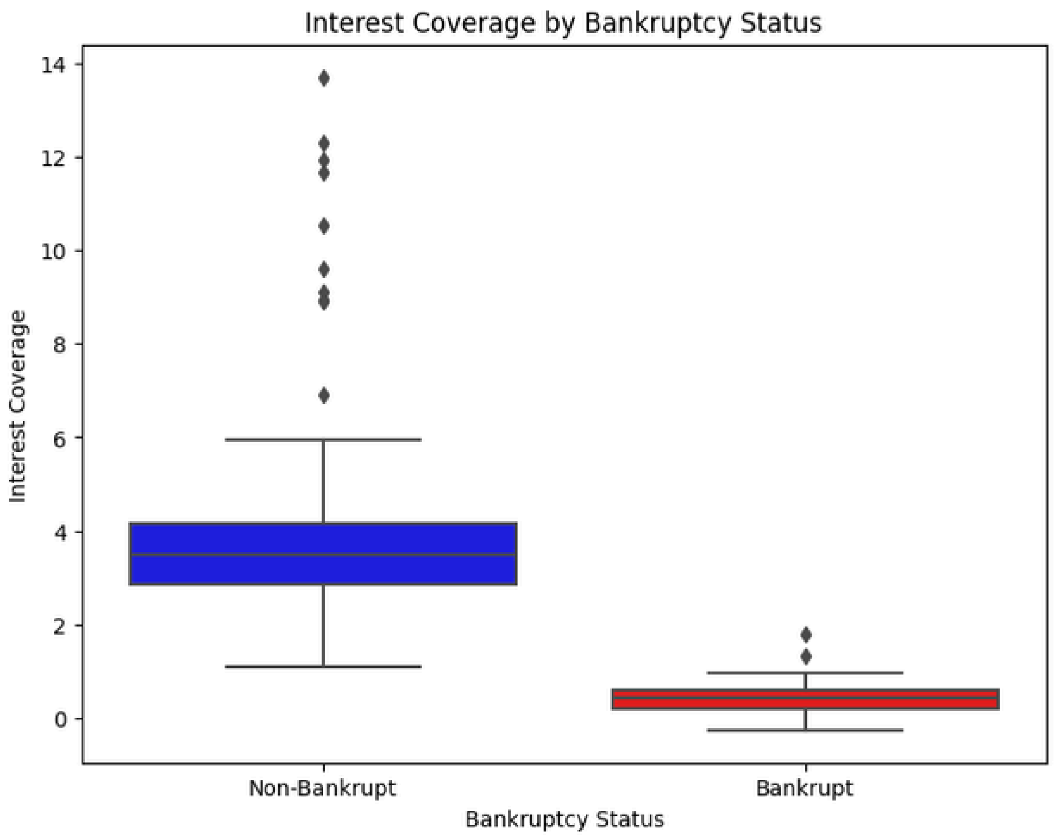
Business hasn't built cash reserves over time

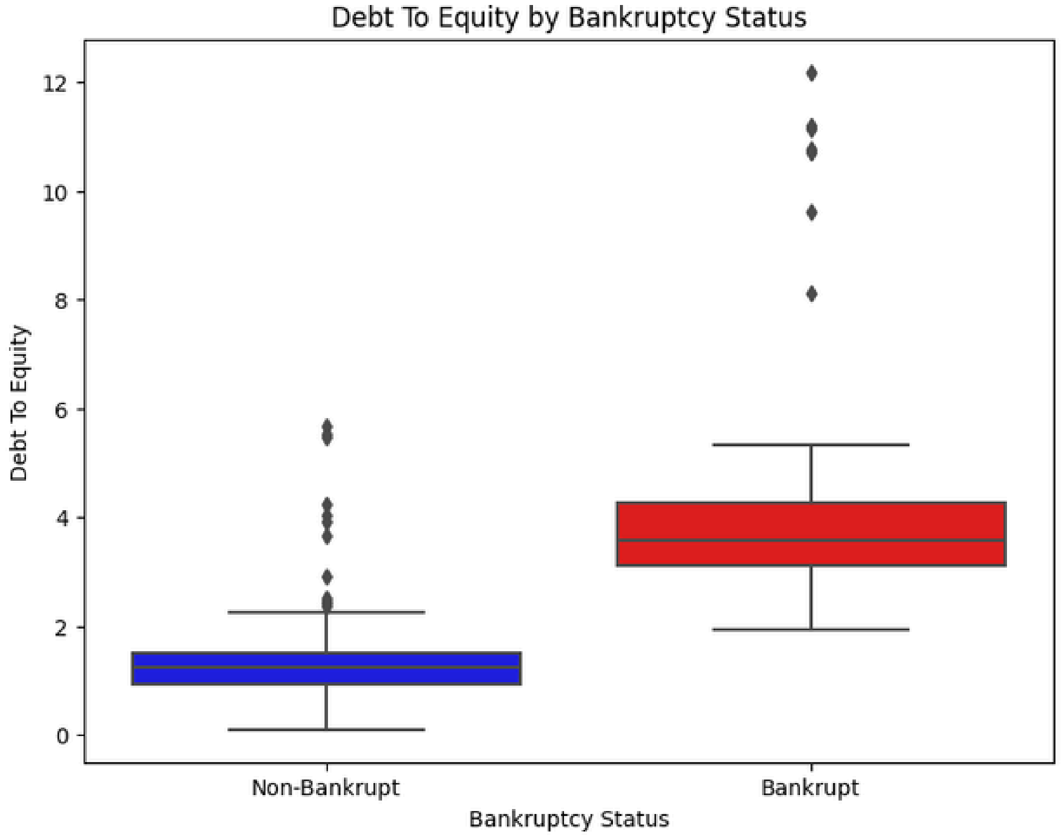




- Low liquidity weakens financial strength and limits the firm's ability to fund operations internally.
- Low retained earnings mean weak reserves, making the company vulnerable to shocks.
- As cash tightens, efficiency drops → ROA falls → profit margins shrink, signalling growing financial distress.

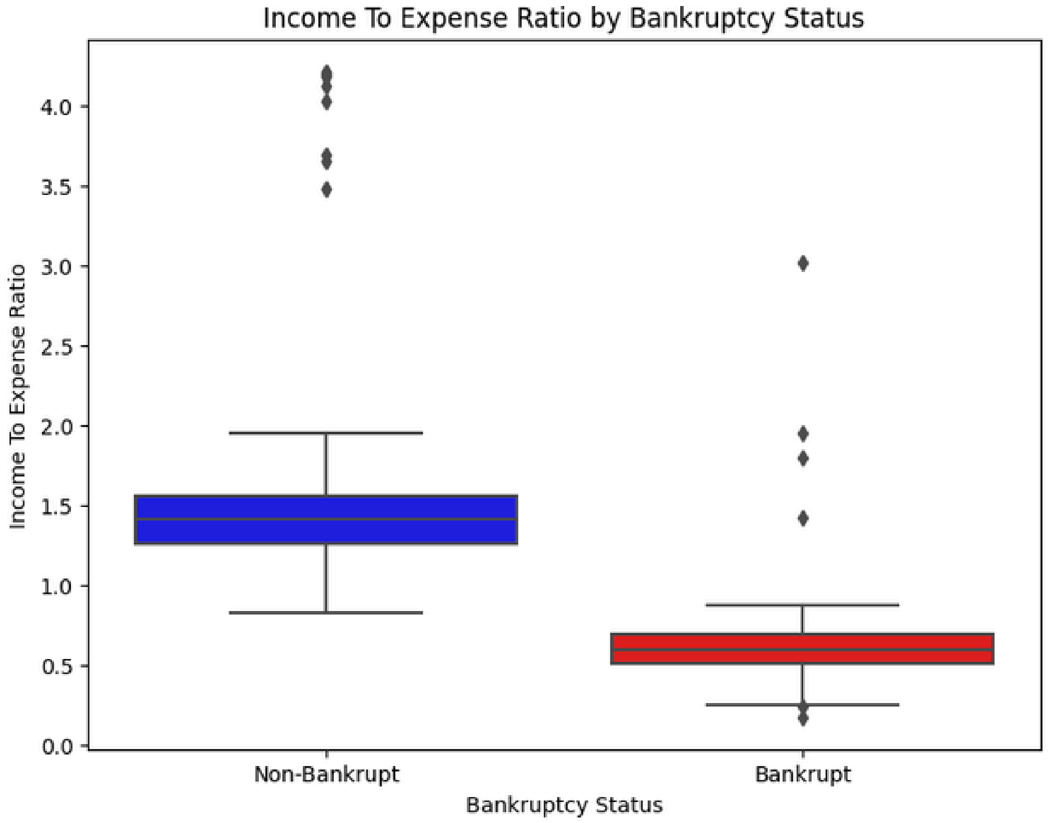
The Point of Bankruptcy





Earnings are no longer enough to pay interest

High Debt to Equity Ratio shows that the bankrupt companies are overleveraged



- As profits decline while debt stays high, the company hits a breaking point
- Interest coverage falls — earnings can no longer cover interest payments
- A high debt-to-equity ratio shows over-leverage, and expenses start exceeding income
- If this situation persists over time, with expenses consistently outpacing profits, bankruptcy is inevitable.

THE PROBLEM AND THE SOLUTION

- Early liquidity stress weakens day-to-day operations from debt, slow receivables, operational inefficiencies, and rising short-term obligations

- Retained earnings were low, profitability dropped, and ROA shrank. Even with revenues coming in, the company couldn't build reserves or improve efficiency

- The persistent imbalance between income and obligations makes it increasingly difficult to meet payments. If this situation continues, the firm is no longer able to sustain operations

Short Term Solution

- Restructure debt, reduce unnecessary expenses, improve cash flow management

Long Term Solution

- Strengthen retained earnings, optimize leverage, improve operational efficiency, build financial reservest

Monitor financial health: Regularly track ratios like ROA, interest coverage, working capital to prevent recurrence.

01. The Problem

02. The Solution

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Samudika

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